



# Using dynamic software to explore linear relationships

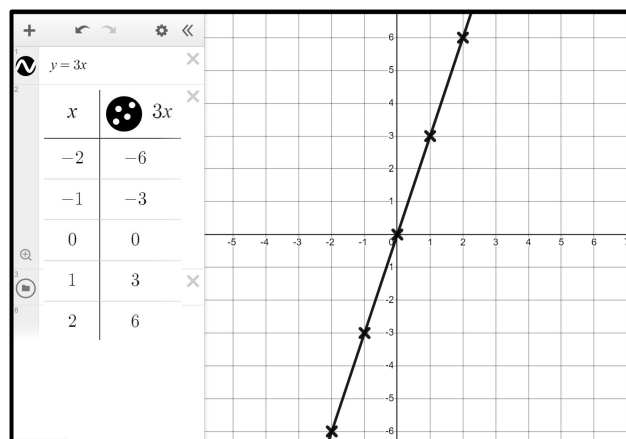
## Task A: Changing the gradient

1) You are the teacher.

Plot the line  $y = 3x$  and explain to our Oak students the link between the algebra, the table and the graph.

*You may have said:*

*The algebra ( $y = 3x$ ) tells us that the  $y$ -coordinate is 3 times the  $x$ -coordinate; the table and coordinates reflect that.*



*In  $y = mx$  the  $m$  represents the gradient. For  $y = 3x$  we see the gradient of 3 on the graph and a change of +3 in  $y$  for every +1 change in  $x$  in the table.*

2) Plot the lines  $y = -4x$  and  $y = \frac{1}{4}x$

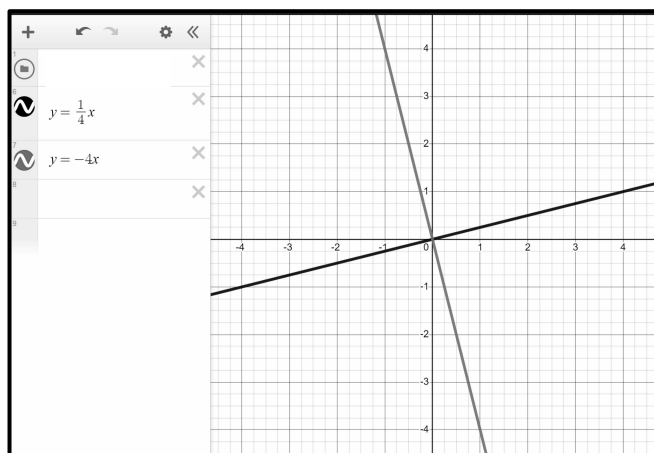
Compare the two lines.

*You may have said:*

*Same  $y$ -intercept of (0,0)*

*One has a positive gradient, one a negative gradient.*

*Respective gradients of -4 and  $\frac{1}{4}$*

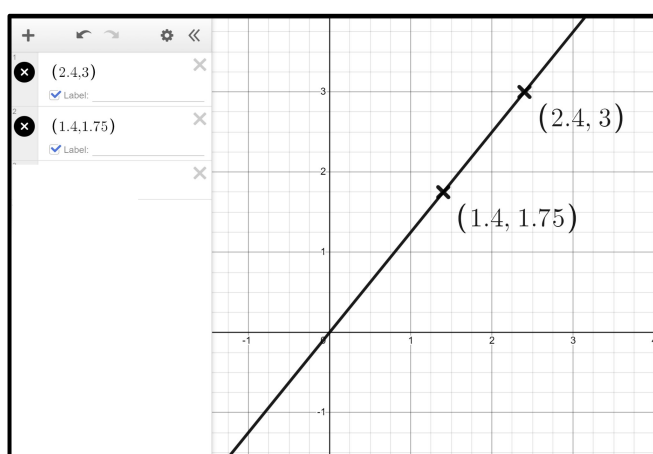


3) Plot the coordinates (2.4, 3) and (1.4, 1.75) and draw the straight line that goes through both and the origin. What is this line?

*Between the coordinates you see a change of +1 in  $x$  and +1.25 change in  $y$  telling us there is a gradient of +1.25*

*Lines in the form of  $y = mx$  go through the origin.*

*With a gradient of +1.25 this must therefore be the line  $y = 1.25x$*



4) Draw the lines that recreate this pattern.

$$y = \frac{1}{4}x \quad y = -\frac{1}{4}x \quad y = x \quad y = -x \quad y = 4x \quad y = -4x$$

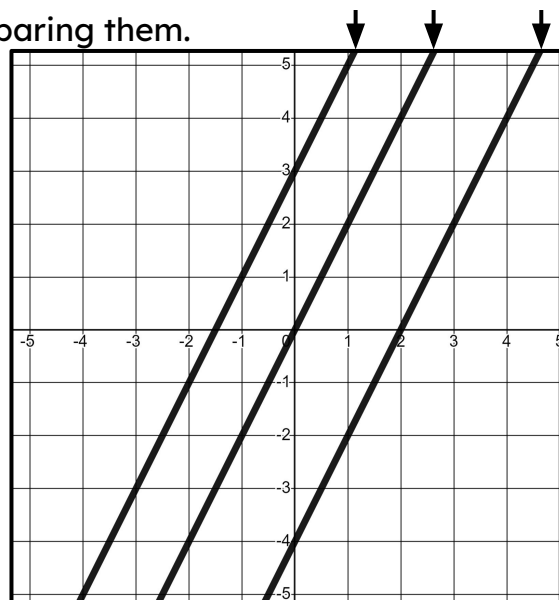
## Task B: Changing the $y$ -intercept

1) Plot the lines and write a sentence or two comparing them.

$y = 2x$  All three lines have the same gradient 2  
This makes them parallel.

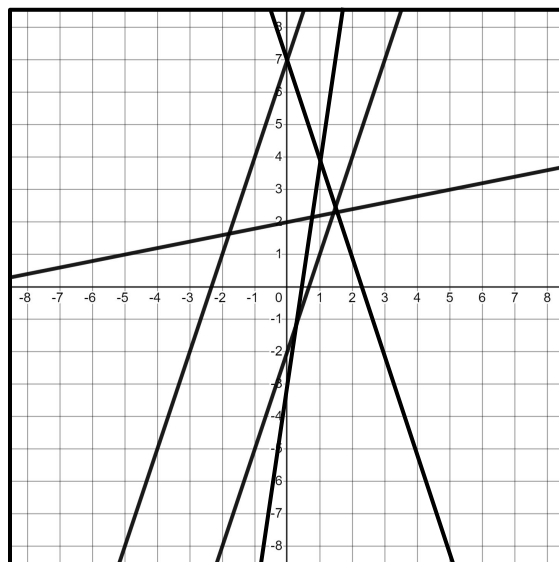
$y = 2x + 3$  Translation of  $y = 2x$  by  $+3$  in the  
 $y$ -direction.  $y$ -intercept of  $(0, 3)$

$y = 2x - 4$  Translation of  $y = 2x$  by  $-4$  in the  
 $y$ -direction.  $y$ -intercept of  $(0, -4)$

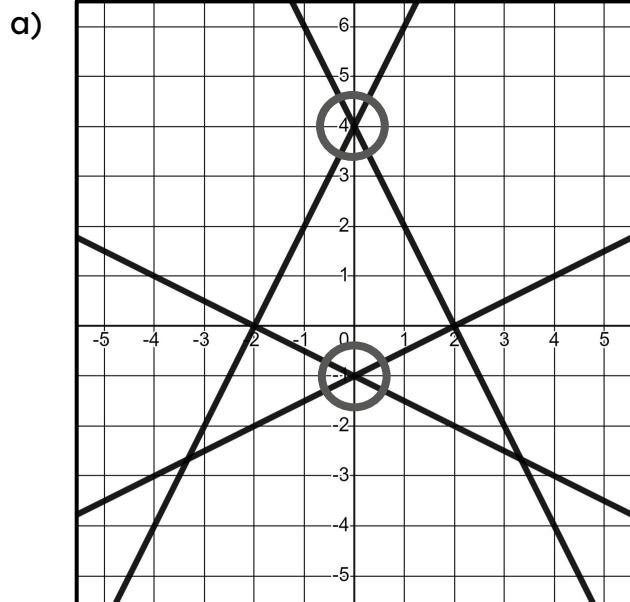


2) Plot the lines and complete the table.

| Line           | Gradient | $y$ -intercept |
|----------------|----------|----------------|
| $y = 3x + 7$   | 3        | $(0, 7)$       |
| $y = 3x - 2$   | 3        | $(0, -2)$      |
| $y = 0.2x + 2$ | 0.2      | $(0, 2)$       |
| $y = 7x - 3$   | 7        | $(0, -3)$      |
| $y = 7 - 3x$   | -3       | $(0, 7)$       |

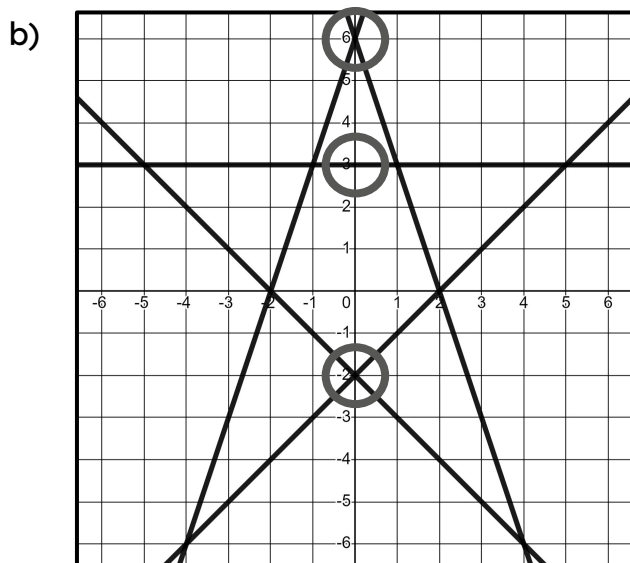


3) Draw the lines that recreate these patterns.



Two lines of  $y$ -intercept  $(0, 4)$ , two lines of  $y$ -intercept  $(0, -1)$

|   |  |                         |  |
|---|--|-------------------------|--|
| 1 |  | $y = \frac{1}{2}x - 1$  |  |
| 2 |  | $y = -1 - \frac{1}{2}x$ |  |
| 3 |  | $y = 2x + 4$            |  |
| 4 |  | $y = 4 - 2x$            |  |



$y$ -intercepts of  $(0, 6)$ ,  $(0, 3)$ , and  $(0, -2)$

|   |  |              |  |
|---|--|--------------|--|
| 1 |  | $y = 3x + 6$ |  |
| 2 |  | $y = 6 - 3x$ |  |
| 3 |  | $y = x - 2$  |  |
| 4 |  | $y = -x - 2$ |  |
| 5 |  | $y = 3$      |  |

